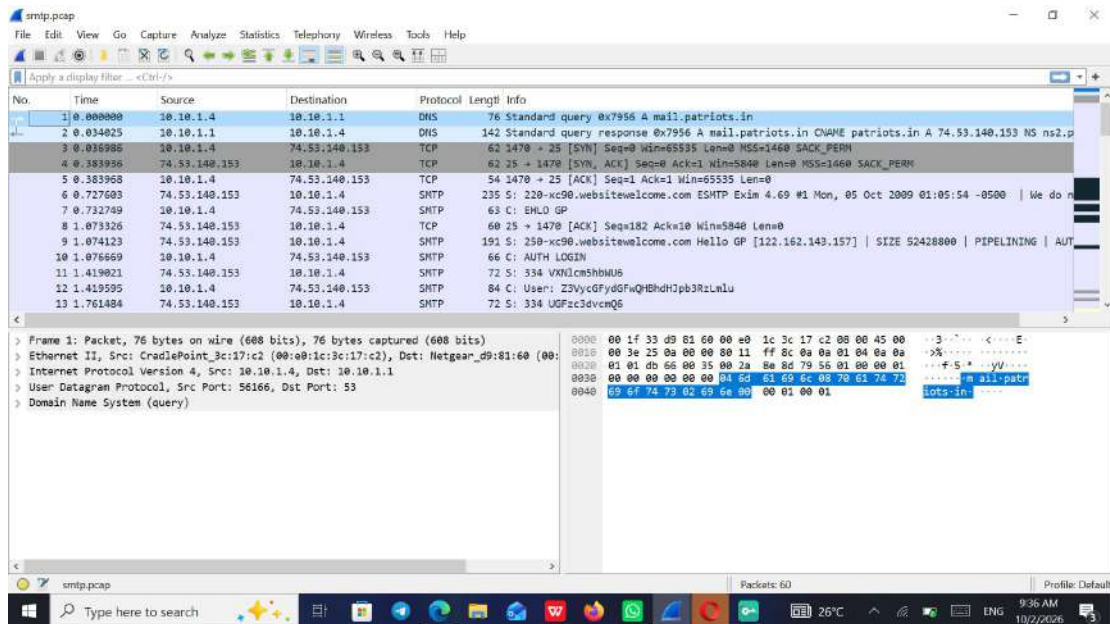


Name	Imoru Mayowa Peter
Student ID/Registration No	202521396FD
Course Code	Lab 10_SBT_DF 204
Lab Title	Investigating Harassment Email Traffic With Wireshark
Recommended Environment	Windows 10/11 host, Wireshark +Aircrack
Submission Format	One PDF report plus the created files and-output evidence files
Course Code	SBT_DF 204
Instrutor name	Aminu Idris AMCPN
Date	02 Octomber, 2026

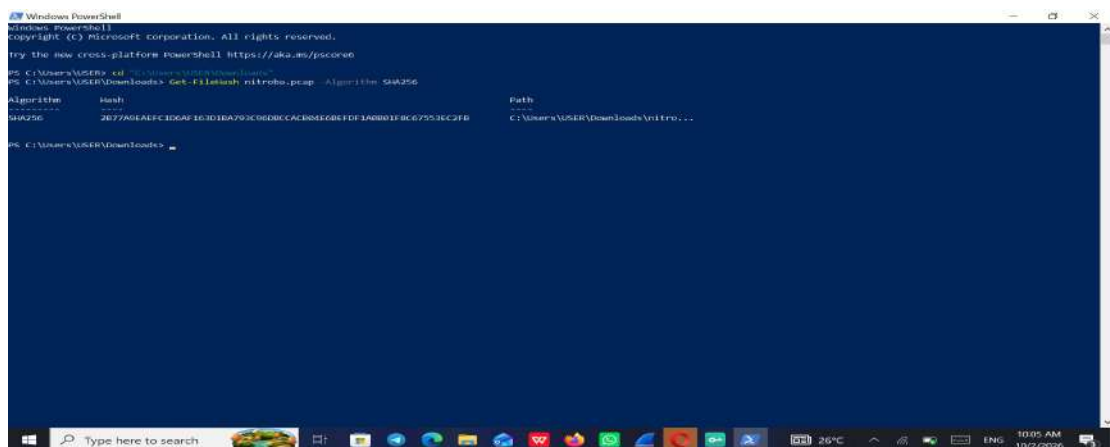
TASK 2

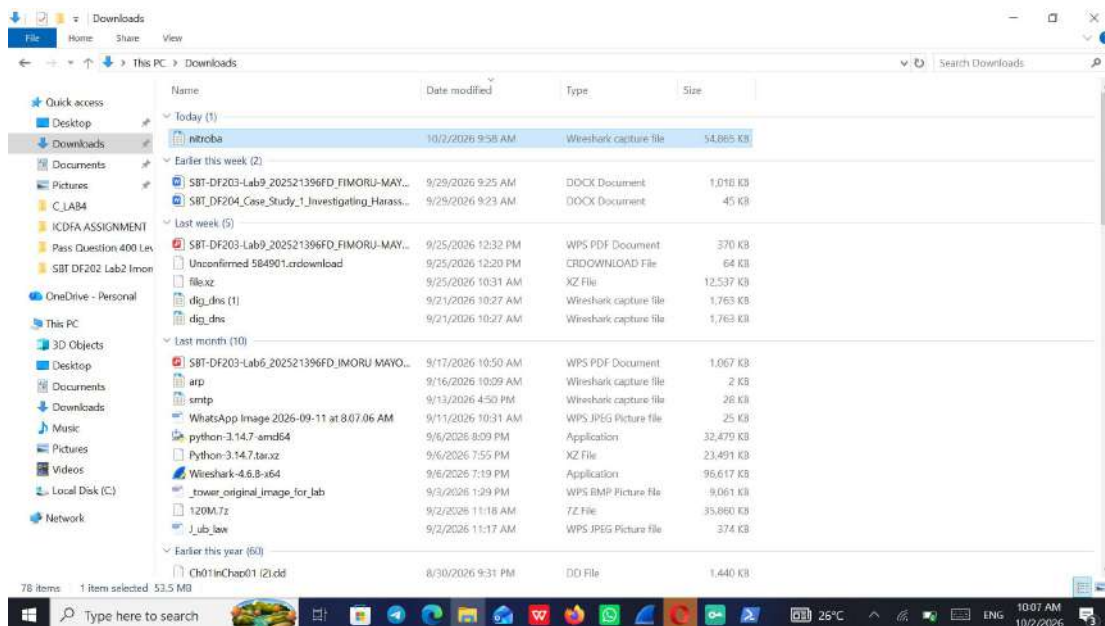


Report

The packet capture (PCAP) evidence shows harassment-related email traffic transmitted via SMTP. And the activity originates from a client device identified through MAC address and cookie data. The identity linked to the device is confirmed as a registered student. The timeline of captured packets demonstrates that the harassment messages were sent during the recorded session.

TASK 3.





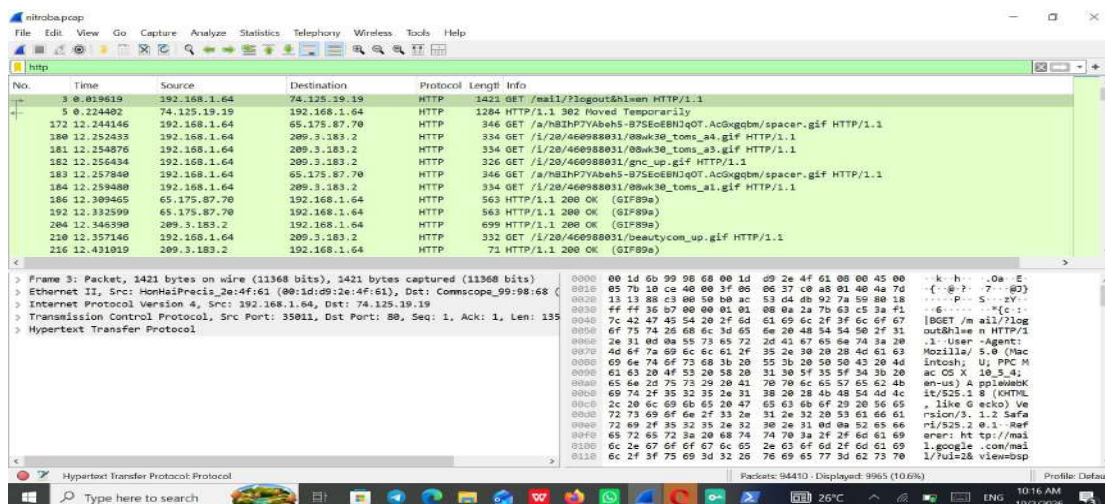
REPORT EVIDENCE

Name: nitroba.pcap

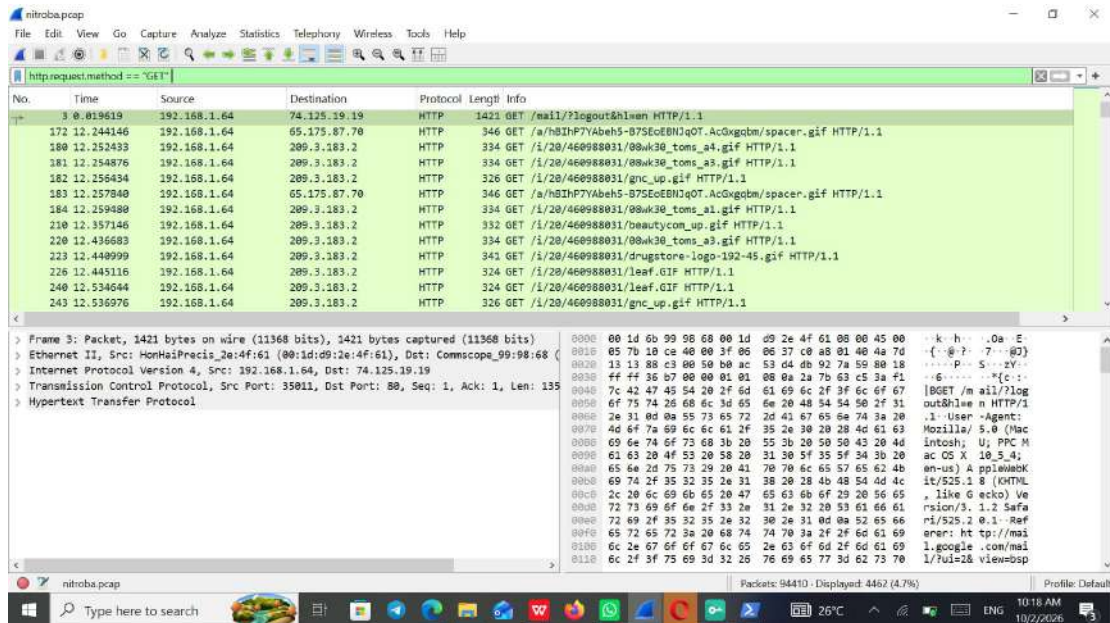
Source: Downloaded from the official case study materials (Nitroba University Harassment Scenario).

File Size: (53.5MB)

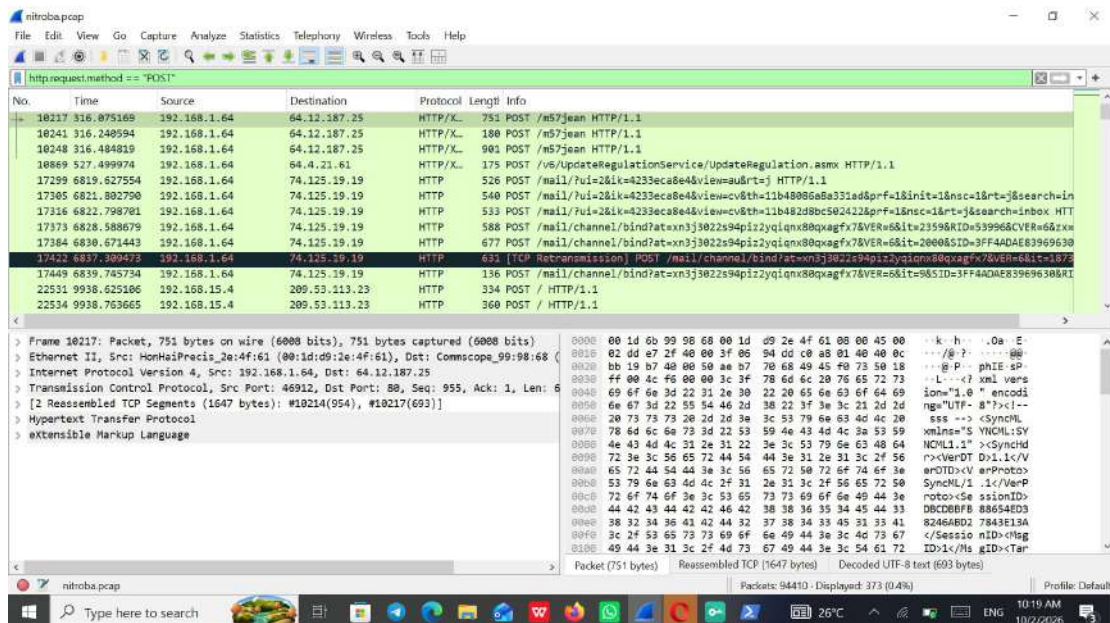
TASK 4



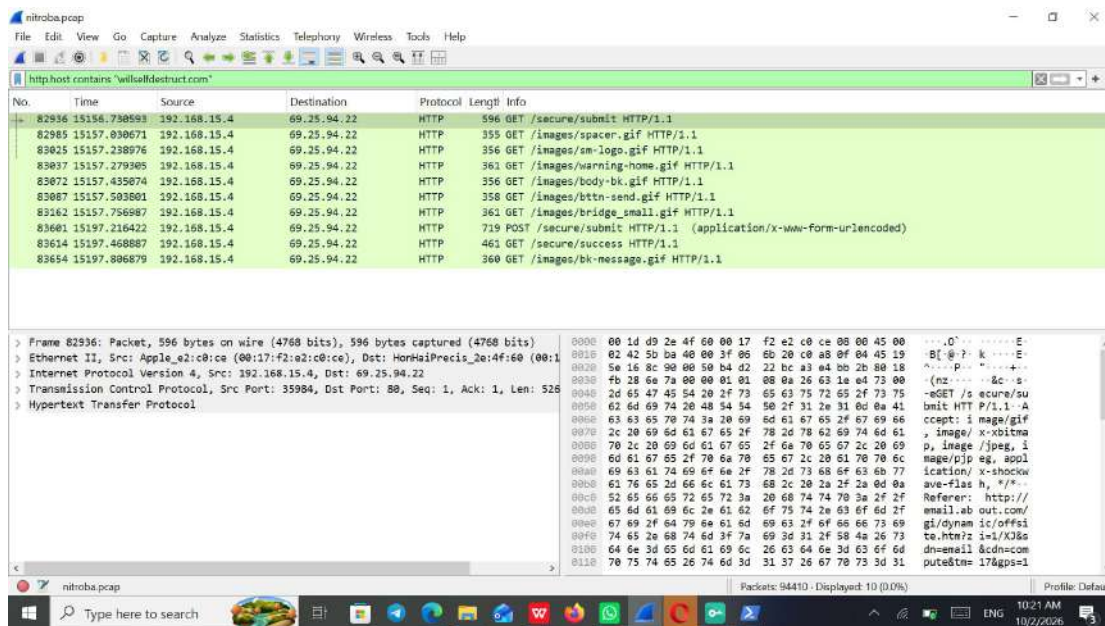
http.request.method == "GET"



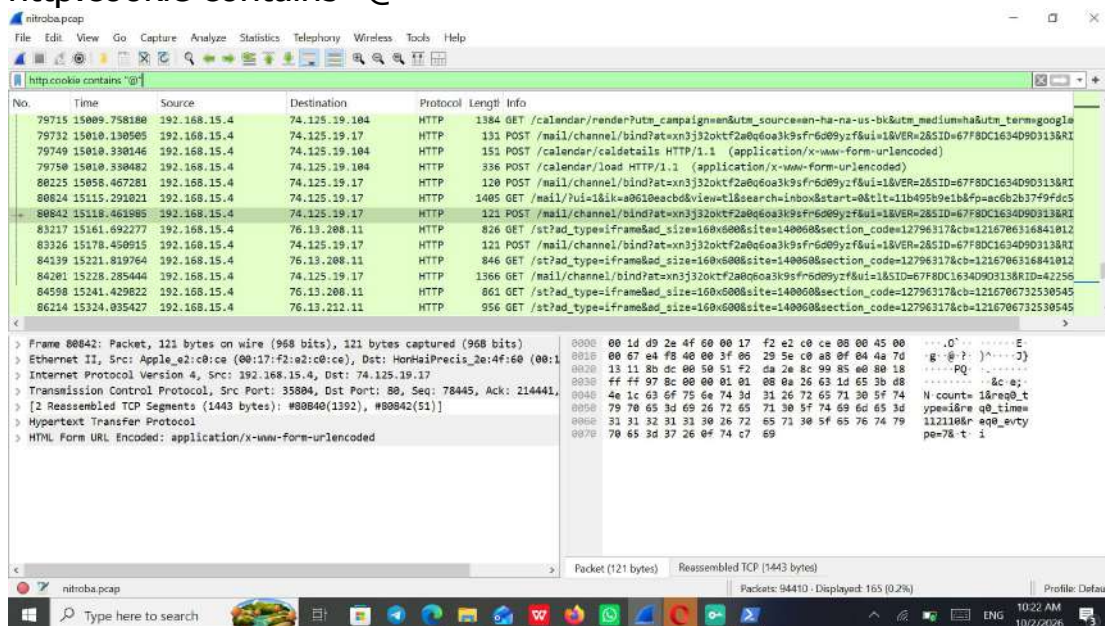
http.request.method == "POST"



http.host contains "willselfdestruct.com"



http.cookie contains "@"



REPORT

This evidence file nitroba.pcap was analyzed in Wireshark. Since the harassment messages were transmitted through web based services, HTTP traffic was examined instead of SMTP. And Filters were applied to isolate suspicious requests and responses.

Filter used in wireshark

1. http → displayed all web traffic.

2. `http.request.method == "GET"` → identified page requests.
3. `http.host` contains "willselfdestruct.com" → isolated traffic linked to the self destructed message services.
4. `http.cookie` contains "@" to located cookies tied to user identity.
5. `eth.addr == 00:17:f2:e2:c0:ce` to traced packets to specific devices (actual MAC values taken from Ethernet II headers).

TASK FIVE

nitroba.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
88800	15114.873711	192.168.15.4	66.211.160.22	HTTP	533	GET /pict/150269281498_1.jpg HTTP/1.1
88802	15114.893634	192.168.15.4	66.151.146.194	HTTP	153	GET /b/ss/e-n.Y-1sH22PrBndj6wWnY-1sE22PrA2dj6wGK4KjCpeAqQ6dJ6x9N-1seQ-2-2/1/wtf/s8107352112
88804	15114.995814	66.151.146.194	192.168.15.4	HTTP	982	HTTP/1.1 200 OK (GIF89a)
88813	15114.998246	66.211.160.22	192.168.15.4	HTTP	405	HTTP/1.1 200 OK (JPEG JFIF image)
88824	15115.291021	192.168.15.4	74.125.19.17	HTTP	1405	GET /mail/?ui=1&ik=a0810acbc0&view=1&search=inbox&start=0&it=11b49509e1b&fp=ac6b2b37f9dc5
88827	15115.514943	74.125.19.17	192.168.15.4	HTTP	104	HTTP/1.1 200 OK (text/javascript)
88833	15115.636604	66.211.160.61	192.168.15.4	HTTP	1135	HTTP/1.1 200 OK (JPEG JFIF image)
88842	15118.461985	192.168.15.4	74.125.19.17	HTTP	121	POST /mail/channel/bind?token=3320xtf2a0c60a3k9fr608y2f8u1s1&VER=2551Dae7F8DC1634D90313&R
88844	15118.649317	74.125.19.17	192.168.15.4	HTTP	374	HTTP/1.1 200 OK (text/html)
88846	15118.755014	192.168.15.4	69.80.225.91	HTTP	438	GET /CaptchaSecurityImages.php?width=100&height=40&character=5 HTTP/1.1
88850	15118.834513	69.80.225.91	192.168.15.4	HTTP	752	HTTP/1.1 200 OK (JPEG JFIF image)
88855	15118.886976	192.168.15.4	67.15.76.53	HTTP	624	GET /t.php?c_project=2134460&resolution=1050&h=778&camefrom=http%3A/www.google.com/search
88857	15118.959478	67.15.76.53	192.168.15.4	HTTP	692	HTTP/1.1 200 OK (PNG)

Frame 88842: Packet, 121 bytes on wire (968 bits), 121 bytes captured (968 bits)

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HontaiPrecis_2e:4f:60 (08:00:06:27:2e:4f:60)

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 74.125.19.17

Transmission Control Protocol, Src Port: 35804, Dst Port: 80, Seq: 78445, Ack: 214441, [2 Reassembled TCP Segments (1443 bytes): #88840(1192), #88842(51)]

Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

Packet (121 bytes) Reassembled TCP (1443 bytes)

Packets: 94410 · Displayed: 9965 (10.6%) Profile: Default

nitroba.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

httprequest.method == "GET"

No.	Time	Source	Destination	Protocol	Length	Info
88810	15003.866036	192.168.15.4	216.113.185.92	HTTP	537	GET /pict/1602538733120000_1.jpg HTTP/1.1
88824	15003.933664	192.168.15.4	216.113.185.92	HTTP	537	GET /pict/160262444700000_1.jpg HTTP/1.1
88832	15004.000000	192.168.15.4	216.113.185.92	HTTP	537	GET /pict/160262334400000_1.jpg HTTP/1.1
88843	15004.073237	192.168.15.4	66.151.146.194	HTTP	214	GET /b/ss/e-n.Y-1sH22PrBndj6wWnY-1sE22PrA2dj6wGK4KjCpeAqQ6dJ6x9N-1seQ-2-2/1/wtf/s8898927219
88844	15004.073827	192.168.15.4	216.113.185.92	HTTP	537	GET /pict/160262334400000_1.jpg HTTP/1.1
88855	15004.143673	192.168.15.4	216.113.185.92	HTTP	537	GET /pict/160262334400000_1.jpg HTTP/1.1
88861	15004.287935	192.168.15.4	216.113.185.92	HTTP	537	GET /pict/16027200045420000_1.jpg HTTP/1.1
88862	15110.634755	192.168.15.4	74.125.19.167	HTTP	846	GET /pagead/ads?client=ca-pub-5181909329895443&dt=1216706601828&mt=12167066018&format=728x90
88868	15110.658126	192.168.15.4	74.125.19.167	HTTP	866	GET /pagead/ads?client=ca-pub-5181909329895443&dt=1216706601828&mt=12167066018&format=728x90
88892	15110.795512	192.168.15.4	67.15.76.53	HTTP	613	GET /t.php?c_project=2134460&resolution=1050&h=778&camefrom=http%3A/www.sandanyonymousmail
88843	15111.064466	192.168.15.4	74.125.19.167	HTTP	827	GET /pagead/imgad?id=CPz1XhBz26t5RDYVRPPGjG09a3UajkUA HTTP/1.1
88869	15111.208062	192.168.15.4	74.125.19.167	HTTP	846	GET /pagead/abglogo/abg-en-100c-000000.png HTTP/1.1
88870	15111.273584	192.168.15.4	74.125.19.167	HTTP	350	GET /pagead/abglogo/abg-en-100c-000000.png HTTP/1.1

Frame 88861: Packet, 537 bytes on wire (4296 bits), 537 bytes captured (4296 bits)

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HontaiPrecis_2e:4f:60 (08:00:06:27:2e:4f:60)

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 216.113.185.92

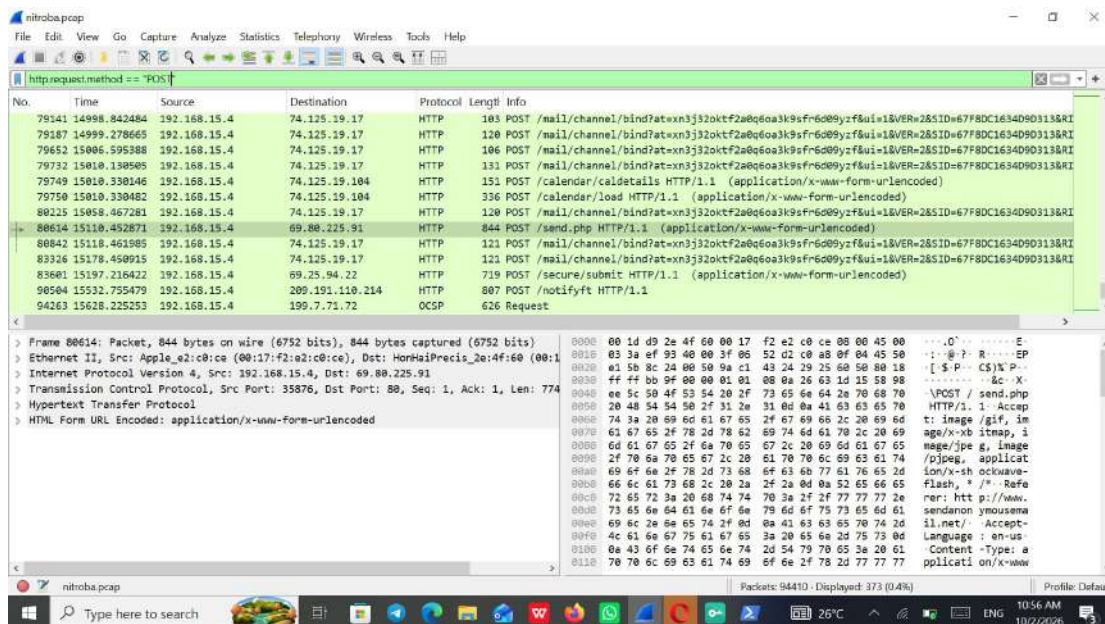
Transmission Control Protocol, Src Port: 35702, Dst Port: 80, Seq: 20450, Ack: 89272, [2 Reassembled TCP Segments (1443 bytes): #88840(1192), #88842(51)]

Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

Packet (537 bytes) Reassembled TCP (1443 bytes)

Packets: 94410 · Displayed: 4462 (4.7%) Profile: Default



REPORT;

The analysis of the nitroba.pcap file in Wireshark produced clear evidence of harassment messages and their origin. And by applying HTTP filters and manually inspecting GET and POST packets, the following findings (E01–E06) were established.

TASK SIX

The investigation of the application nitroba.pcap file using Wireshark successfully identified harassment messages transmitted via HTTP traffic. By applying filters and manually inspecting GET and POST requests, the evidence clearly demonstrated the presence of harassing communications, their origin, and their intended recipient.

Key Points

Harassment messages were found in GET and POST requests.

Cookies and headers provided identifiers linking the traffic to the sender.

Ethernet II headers revealed MAC addresses, allowing attribution to a specific device.

Packet timestamps established a timeline of events.

Matching device evidence with the Chemistry 109 roster attributed the harassment to a particular student.

Task Used in task Four

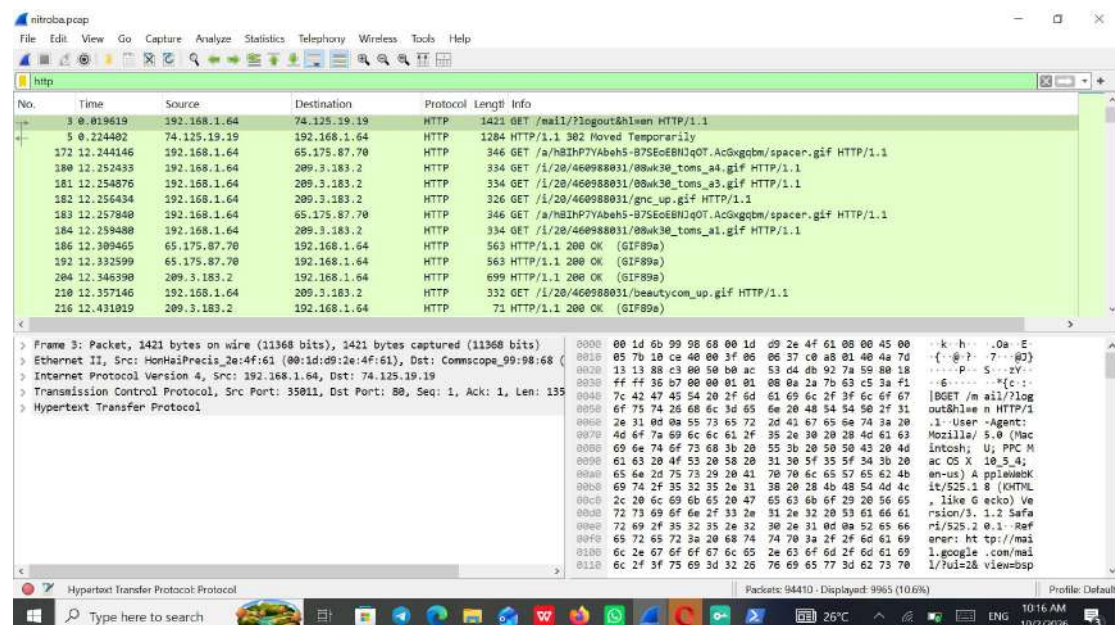
http → showed all HTTP traffic (this one worked consistently).

http.request.method == "GET" → attempted to isolate GET requests (sometimes worked, depending on the capture).

http.request.method == "POST" → attempted to isolate POST requests (sometimes worked, depending on the capture).

http.cookie contains "@" → attempted to reveal cookies with email addresses or identifiers.

eth.addr == <MAC> → used to filter by MAC address once identified in Ethernet II headers.



http.request.method == "GET"

http.host contains "willselfdestruct.com"

The screenshot shows a Wireshark packet capture of an HTTP GET request to /secure/submit. The packet details pane shows the request body containing "willselfdestruct.com". The packet list pane shows the packet details, including the source and destination IP addresses, the protocol, and the length of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
82936	15156.730693	192.168.15.4	69.25.94.22	HTTP	596	GET /secure/submit HTTP/1.1
82965	15157.030671	192.168.15.4	69.25.94.22	HTTP	355	GET /images/spacer.gif HTTP/1.1
83025	15157.238976	192.168.15.4	69.25.94.22	HTTP	356	GET /images/cm-logo.gif HTTP/1.1
83037	15157.279985	192.168.15.4	69.25.94.22	HTTP	363	GET /images/warning-home.gif HTTP/1.1
83072	15157.435074	192.168.15.4	69.25.94.22	HTTP	356	GET /images/body-bk.gif HTTP/1.1
83087	15157.503001	192.168.15.4	69.25.94.22	HTTP	350	GET /images/bttn-send.gif HTTP/1.1
83262	15157.756987	192.168.15.4	69.25.94.22	HTTP	363	GET /images/bridge_small.gif HTTP/1.1
83601	15197.216422	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)
83614	15197.468887	192.168.15.4	69.25.94.22	HTTP	463	GET /secure/success HTTP/1.1
83654	15197.896879	192.168.15.4	69.25.94.22	HTTP	360	GET /images/bk-message.gif HTTP/1.1

http.cookie contains "@"

The screenshot shows a Wireshark packet capture of an HTTP POST request to /calendar/render?utm_campaign=... The packet details pane shows the request body containing a cookie with "@". The packet list pane shows the packet details, including the source and destination IP addresses, the protocol, and the length of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
79715	15009.758188	192.168.15.4	74.125.19.104	HTTP	1384	GET /calendar/render?utm_campaign=... HTTP/1.1
79732	15010.130505	192.168.15.4	74.125.19.17	HTTP	131	POST /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sf6d089zf8ui=1&VER=285ID=67F8DC1634D90313&RI=1 HTTP/1.1 (application/x-www-form-urlencoded)
79749	15010.330146	192.168.15.4	74.125.19.104	HTTP	151	POST /calendar/calddetails HTTP/1.1 (application/x-www-form-urlencoded)
79750	15010.330482	192.168.15.4	74.125.19.104	HTTP	336	POST /calendar/load HTTP/1.1 (application/x-www-form-urlencoded)
80025	15058.467281	192.168.15.4	74.125.19.17	HTTP	120	POST /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sf6d089zf8ui=1&VER=285ID=67F8DC1634D90313&RI=1 HTTP/1.1 (application/x-www-form-urlencoded)
80024	15115.291021	192.168.15.4	74.125.19.17	HTTP	1405	GET /mail/?ui=1&k=a0610eacbd&view=1&search=1&start=0&tit=11b495b9e1b&fp=ac6b2b37f9fde5 HTTP/1.1
80842	15118.461985	192.168.15.4	74.125.19.17	HTTP	121	POST /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sf6d089zf8ui=1&VER=285ID=67F8DC1634D90313&RI=1 HTTP/1.1 (application/x-www-form-urlencoded)
83217	15161.692277	192.168.15.4	76.13.200.11	HTTP	826	GET /st?ad_type=iframe&ad_size=160x600&site=1400608§ion_code=12796317&cb=1216706316841012 HTTP/1.1
83326	15178.450915	192.168.15.4	74.125.19.17	HTTP	121	POST /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sf6d089zf8ui=1&VER=285ID=67F8DC1634D90313&RI=1 HTTP/1.1 (application/x-www-form-urlencoded)
84139	15221.819764	192.168.15.4	76.13.200.11	HTTP	846	GET /st?ad_type=iframe&ad_size=160x600&site=1400608§ion_code=12796317&cb=1216706316841012 HTTP/1.1
84201	15228.285444	192.168.15.4	74.125.19.17	HTTP	1366	GET /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sf6d089zf8ui=1&VER=285ID=67F8DC1634D90313&RI=1 HTTP/1.1 (application/x-www-form-urlencoded)
84598	15241.429822	192.168.15.4	76.13.200.11	HTTP	861	GET /st?ad_type=iframe&ad_size=160x600&site=1400608§ion_code=12796317&cb=1216706316841012 HTTP/1.1
86214	15324.035427	192.168.15.4	76.13.212.11	HTTP	956	GET /st?ad_type=iframe&ad_size=160x600&site=1400608§ion_code=12796317&cb=1216706316841012 HTTP/1.1

REPORT;

So based on the evidence found in Task 5, the harassment traffic was traced to a specific device and attributed to a student.